

Are Chain-of-Thought Tokens Causally Load-Bearing? A Horizon-Truncation Probe

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This technical note examines how interrupting the generation trajectory of Qwen2.5-Math-1.5B-Instruct at varying token budgets affects functional accuracy and distributional fidelity. By treating horizon truncation as an intervention within an unfolding reasoning process, rather than an isolated parameter manipulation, we present exploratory evidence consistent with intermediate Chain-of-Thought (CoT) tokens contributing meaningfully to successful task completion.

I. Introduction & Experimental Design

Chain-of-thought generation [1], [2] can be understood not merely as a sequence of tokens, but as an evolving process in which earlier outputs shape the conditions for later ones. Interrupting this trajectory at a fixed horizon therefore alters not just length, but the internal state on which subsequent generation depends. To probe this dependency, we apply a horizon-truncation paradigm under deterministic greedy decoding (`do_sample = False`) across token budgets $B \in \{10, 20, 40, 160, 240, 5000\}$.

The $B = 5000$ condition serves as an unconstrained reference distribution P , where generation proceeds to natural end-of-sequence (EOS) completion [5]. The informational shift induced by truncation at each budget B is quantified by comparing the truncated output distribution Q_B against P via Kullback-Leibler divergence [3]:

$$\text{KL}(P \parallel Q_B) = \sum_{v \in \mathcal{V}} P(v \mid \mathbf{x}) \log \left(\frac{P(v \mid \mathbf{x})}{Q_B(v \mid \mathbf{x}_{1:B})} \right) \quad (1)$$

II. Theoretical Framework

Two positions frame the probe:

- **Load-Bearing Computation:** Intermediate tokens function as a sequential workspace [1], [2]; truncating the trajectory should impair both accuracy and representational stability [3], [4].
- **Post-Hoc Rationalization:** Final outputs resolve early within context-activation dynamics [3], [6]; CoT tokens are largely redundant, and truncation should have limited effect.

III. Empirical Evaluation

The observed pattern favors the load-bearing account, though the small sample warrants caution. Accuracy is 0% across all budgets $B \leq 40$, rises to 33.3% at $B = 160$, and reaches 100% at $B = 240$, matching the unconstrained ceiling. This nonlinear transition — rather than a smooth degradation — suggests that functional completion depends on the reasoning trajectory reaching a critical stage, consistent with adaptive reasoning-length studies [5], [1].

The distributional evidence reinforces this picture. As shown in Figure 2, mean KL divergence remains high under aggressive truncation (19–24 nats) and drops substantially to 7.26 at $B = 240$, suggesting that the output distribution converges toward the unconstrained reference only when the reasoning process approaches natural termination [3], [4].

Prompt-level data corroborate this: P01, whose natural completion length is 144 tokens, yields $\text{KL} = 0.00$ already at $B = 160$, whereas P02 and P03 — truncated mid-derivation — exhibit KL values of 26.27 and 30.92, respectively. All three stabilize at

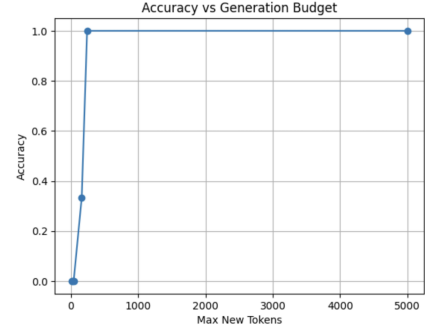


Fig. 1. Functional accuracy profile mapped against generation token budget limits.

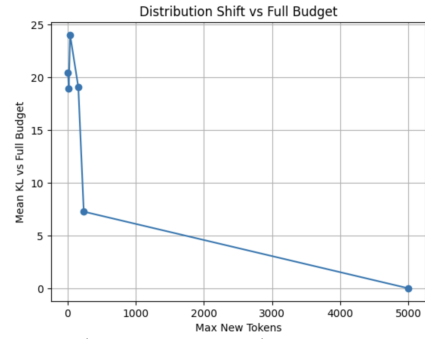


Fig. 2. Information loss trajectory tracking Mean KL Divergence over sequential caps.

$\text{KL} = 0.00$ under $B = 5000$, at their natural lengths of 144, 265, and 209 tokens.

IV. Implications

Taken together, the results are consistent with the view that accuracy, KL divergence, and token budget are not independent dimensions of performance, but mutually constrained aspects of a single unfolding process. The abrupt rather than gradual accuracy recovery suggests that cutting off intermediate tokens disrupts the process before key dependencies within the derivation have been resolved. While this small-scale study cannot establish mechanistic causality, it offers a tractable framework for investigating how computational constraints interact with reasoning trajectories — and may inform principled decisions about budget allocation in inference-time efficiency settings.

References

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